VEGABrain: a Visualizer of Enriched Gene Analysis for the developing human brain

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Abstract

RNA sequencing datasets are widely reused resource in developmental neuroscience, yet inspecting it spatially remains unclear and difficult. Here, we present VEGABrain, an openly accessible web tool that renders bulk RNA-seq onto brain cortical and subcortical atlas surfaces across five human developmental windows spanning the first trimester to adulthood. Beyond single-gene maps, VEGABrain contributes a composite per-region enrichment layer using single-sample gene set enrichment analysis (ssGSEA) scores from precomputed for 19000 gene sets drawn from Gene Ontology, Reactome, KEGG, BioCarta, the Human Phenotype Ontology and the MSigDB hallmark collection, and aggregated to a score for every gene set, developmental window and brain region. Furthermore, VEGABrain allows users to submit a custom gene list and obtain its ssGSEA map across development, computed on demand. Therefore, VEGABrain serves as a containerised R API offering easy to query expression throughout development and brain areas.

Keywords: Neuroinformatics, Brain development, BrainSpan, Gene set enrichment, ssGSEA, Data visualization, Web application

1 Introduction

The transcriptome of the developing human brain is one of the most heavily reused datasets in neuroscience. The BrainSpan atlas (BrainSpan 2011; Kang et al. 2011; Miller et al. 2014) sampled bulk tissue from cortical and subcortical structures across prenatal and postnatal donors, and subsequent integrative work built on it to relate developmental expression to neuropsychiatric risk (Li et al. 2018) and to evolutionary divergence between human and macaque (Zhu et al. 2018).

What has not kept pace is the ease of *seeing* it. A researcher who wants to know where and when a gene, or a set of genes, is expressed in the developing brain has essentially two options today, and both impose costs that are unrelated to the biological question.

The first is the resource portal. Portals in this family—BrainSpan itself, and the wider family of Allen Institute browsers (Lau et al. 2008; Sunkin et al. 2012)—return the data as tables, line plots or heat maps indexed by structure acronym. This is faithful but not anatomical: the reader must hold the mapping from `DFC`, `MFC` or `STC` to a location in the brain in their head, and a heat map of 26 structure codes does not convey a spatial gradient the way a rendered cortex does. It is also gene-at-a-time. There is no route from a gene set, a pathway, or the output of a differential-expression analysis to a spatial view.

One portal in this family goes considerably further. BrainScope (Huisman et al. 2017) places the Allen adult and developing atlases behind an interactive interface and does accept a user-supplied gene set. Its organising abstraction, however, is a pair of linked t-SNE maps—one of genes, one of samples—in which anatomy is recovered indirectly, through choropleths over user-selected slices that localise the samples; a submitted set is summarised by its mean expression, and functional interpretation is handed off to external enrichment servers. The spatial question it answers is thus one of similarity structure among samples rather than of position in a named parcellation, and it is answered without a composite enrichment score.

The second option is to do it yourself. Excellent programmatic brain-plotting tools exist—`ggseg` and `ggseg3d` in R (Mowinckel and Vidal-Piñeiro 2020), `neuromaps` in Python (Markello et al. 2022), and `abagen` for the adult Allen Human Brain Atlas (Hawrylycz et al. 2012; Markello et al. 2021), and `cerebroViz` for region-level spatiotemporal data (Bahl et al. 2017)—and the case for producing figures programmatically rather than through graphical interfaces has been made convincingly (Chopra et al. 2023): it is reproducible, scriptable and version-controllable. But that route presupposes a user who writes R or Python, who is willing to download and hold a $52{,}376 \times 524$ expression matrix locally, and who will construct and validate the mapping from BrainSpan's tissue nomenclature onto an atlas parcellation. Each of those steps is a place where a project stalls or a mapping silently diverges between laboratories. The barrier is not conceptual; it is logistical, and it falls hardest on exactly the clinicians and experimental biologists whose domain knowledge would make the interpretation valuable.

There is a third gap that neither option addresses. Modern analyses rarely end with one gene. They end with a list—a differentially expressed set, a module from a co-expression network, a panel of risk genes for a disorder. The natural question is then not "where is gene X expressed?" but "where and when is this *programme* active?". Answering it requires collapsing many genes into a single per-sample summary. Single-sample gene set enrichment analysis (ssGSEA) (Barbie et al. 2009), a per-sample variant of GSEA (Subramanian et al. 2005) implemented alongside GSVA

(Hänzelmann et al. 2013), provides exactly such a composite score, and has been used as a quantitative readout of pathway activity in expression data (Yi et al. 2020). To our knowledge no tool has applied it as a *spatial* layer over the developing human brain.

VEGABrain (Visualizer of Enriched Gene Analysis) addresses these three gaps in one interface, reachable at https://mcblab.github.io/BrainMap/ with no installation, account or download. Its contributions are:

1. **Anatomical rendering of BrainSpan across development.** BrainSpan samples are mapped onto Desikan–Killiany cortical parcels (Desikan et al. 2006) and `aseg` subcortical structures (Fischl et al. 2002) through `ggseg` (Mowinckel and Vidal-Piñeiro 2020), and displayed as a lateral/medial/sagittal/coronal panel faceted over five developmental windows. The mapping is published as a single auditable CSV rather than reimplemented per project.
2. **A composite per-region enrichment layer.** ssGSEA scores were precomputed for 18,650 gene sets spanning the Gene Ontology (Ashburner et al. 2000; The Gene Ontology Consortium et al. 2023), Reactome (Gillespie et al. 2022), KEGG (Kanehisa 2000), BioCarta, the Human Phenotype Ontology (Köhler et al. 2021) and the MSigDB hallmark collection (Liberzon et al. 2011, 2015), then aggregated to a mean score per gene set, developmental window and region. Selecting an ontology term yields its spatiotemporal activity map immediately.
3. **On-demand scoring of user-submitted gene lists.** A user may paste an arbitrary list of HGNC symbols; VEGABrain validates them against the matrix, runs ssGSEA for that set on the fly, and returns the same developmental map. This closes the loop from an enrichment result (Chen et al. 2013; Kuleshov et al. 2016) back to anatomy.

Both the figures and the numbers behind them are exportable—SVG for the maps, CSV for the per-region tables—so that VEGABrain is a starting point for downstream analysis rather than a terminal display, consistent with the FAIR principles for reusable research data (Wilkinson et al. 2016). The remainder of this paper describes the data preparation and scoring (Section 2), the tool itself (Section 3), its validation and its position relative to comparable software (Section 4), following the expectations for software papers in this journal (De Schutter et al. 2009).

2 Materials and Methods

2.1 Expression data and preprocessing

VEGABrain is built on the BrainSpan Developmental Transcriptome RNA-Seq release, Gencode v10 RPKM values summarised to genes (BrainSpan 2011; Miller et al. 2014). The distributed matrix contains 52,376 gene rows and 524 sample columns. Rows with duplicate HGNC symbols are dropped, keeping the first occurrence, and values are transformed as $\log_2(\text{RPKM}+1)$. No further normalisation or batch correction is applied; the released RPKM values are used as the provider distributes them, so that scores computed here remain comparable with analyses performed directly on the public matrix. Preprocessing is performed once, offline, by `preparaDados.R`, which writes the objects the running service loads at start-up.

2.2 Developmental windows

BrainSpan donors span 31 distinct ages from 8 post-conceptional weeks to 40 years. Individual ages are too sparsely sampled to render as separate maps, so donors are binned into five developmental windows: first trimester ($n = 5$ donors), second trimester ($n = 10$), third trimester ($n = 5$), infant ($n = 8$) and adult ($n = 14$). The bin boundaries and their donor counts are given in Table 1; the donor count is displayed in every facet label in the interface so that the reader is never shown a map without also being shown how much evidence stands behind it.

2.3 Mapping BrainSpan structures to an atlas

BrainSpan labels tissue by dissected structure (for example *dorsolateral prefrontal cortex*), whereas `ggseg` draws atlas parcels (for example *rostral middle frontal*). Bridging the two is the step that most commonly diverges between reimplementations, so VEGABrain externalises it into a single version-controlled file, `mapeamento_regioes.csv`, with one row per atlas region giving the BrainSpan structure it corresponds to and the macro-region it belongs to. Cortical regions map onto the Desikan–Killiany parcellation (Desikan et al. 2006) and subcortical structures onto the FreeSurfer `aseg` segmentation (Fischl et al. 2002), both accessed through `ggseg` (Mowinckel and Vidal-Piñeiro 2020).

The interface exposes two levels of granularity. At the *micro* scale each of the 42 mapped atlas regions carries its own value. At the *macro* scale regions are pooled into six anatomical groupings—frontal, parietal, temporal and occipital lobes, subcortical structures and cerebellum—the mean is taken within each grouping, and that mean is painted back onto every region belonging to it. The macro scale exists because BrainSpan sampling density varies sharply across structures: pooling trades spatial resolution for donor support, and letting the user switch between the two makes that trade-off explicit rather than hidden. Regions with no corresponding sample are drawn in grey rather than imputed.

Every map is rendered as four stacked panels—right lateral cortex, right medial cortex, subcortical sagittal and subcortical coronal—each faceted into the five developmental windows, giving a single figure in which spatial and temporal structure are read on orthogonal axes. A reference panel renders the same geometry coloured by region name, at either granularity, so that a reader can identify any parcel without leaving the tool.

2.4 Composite enrichment scores

For a gene set G and a sample j, ssGSEA (Barbie et al. 2009) ranks all genes by their expression within that sample and computes an enrichment score from the normalised difference between the empirical cumulative distribution of the genes in G and that of the genes outside it:

$$\mathrm{ES}(G,j) \;=\; \sum_{i=1}^{N}\Big(P_{\mathrm{w}}^{G}(G,j,i) - P^{NG}(G,j,i)\Big), \tag{1}$$

where i indexes genes in decreasing order of expression in sample j, P_{w}^{G} is the weighted cumulative distribution over genes in the set and P^{NG} the cumulative distribution over the rest. Because the score is computed from within-sample ranks, it is invariant to the monotone transformation applied in Section 2.1 and does not require a reference group or a phenotype contrast—which is what makes it usable here, where each sample is a single donor–structure pair rather than a member of a case/control design. We use the `ssgseaParam` interface of the `GSVA` package (Hänzelmann et al. 2013) with default parameters.

Gene sets are retrieved through `msigdbr` (Dolgalev 2025) for *Homo sapiens* and comprise the MSigDB hallmark collection (Liberzon et al. 2015), the canonical pathway subcollections BioCarta, KEGG (Kanehisa 2000) and Reactome (Gillespie et al. 2022), and the full C5 ontology collection, which supplies Gene Ontology biological process, molecular function and cellular component terms (Ashburner et al. 2000; The Gene Ontology Consortium et al. 2023) together with Human Phenotype Ontology terms (Köhler et al. 2021). The resulting 18,650 sets are broken down by collection in Table 1.

Scores are computed once for every gene set against all 524 samples, then aggregated: for each (gene set, developmental window, region) triple the mean score over the contributing samples is stored, and analogously for macro-regions. This reduces roughly 9.8 million sample-level scores to 3,188,466 rows at the micro scale, which is what makes interactive response possible—an ontology query becomes a filter over a precomputed table rather than a fresh enrichment run. When instead a user submits their own gene list, the symbols are validated against the matrix (unmatched symbols are reported back rather than silently dropped), ssGSEA is run for that single set across all samples, and the same aggregation is applied before rendering.

2.5 Implementation and deployment

VEGABrain is split into a stateless computational back end and a static front end (Figure 1). The back end is an R (R Core Team 2026) service built with `plumber` (Schloerke and Allen 2024), which loads the preprocessed objects at start-up and exposes the endpoints listed in Table 2. Maps are rendered with `ggplot2` and `ggseg` (Mowinckel and Vidal-Piñeiro 2020) and serialised as SVG, so that what the browser displays is already a vector figure at publication quality; tabular endpoints return CSV. The front end is a React single-page application served as static files from GitHub Pages, holding no data of its own and calling the API directly. An earlier prototype ran as a single Shiny application (Chang et al. 2025); splitting it into an API and a static client is what lets the front end be hosted at no cost and the compute scale to zero independently of it.

The service is containerised and deployed to Google Cloud Run with 4 GiB of memory and 2 vCPUs. Because the R process is single-threaded, request concurrency is capped at four, and the request timeout is raised to 900 s to accommodate on-demand ssGSEA runs. The service scales to zero between requests, which suits the expected sporadic usage and keeps hosting within a free tier; the front end retries the initial listing calls so that a cold start is absorbed transparently. Both deployments are automated—the front end by a GitHub Actions workflow on push, the API by a single script—and the container definition pins package versions so that a rebuild reproduces the analysis environment.

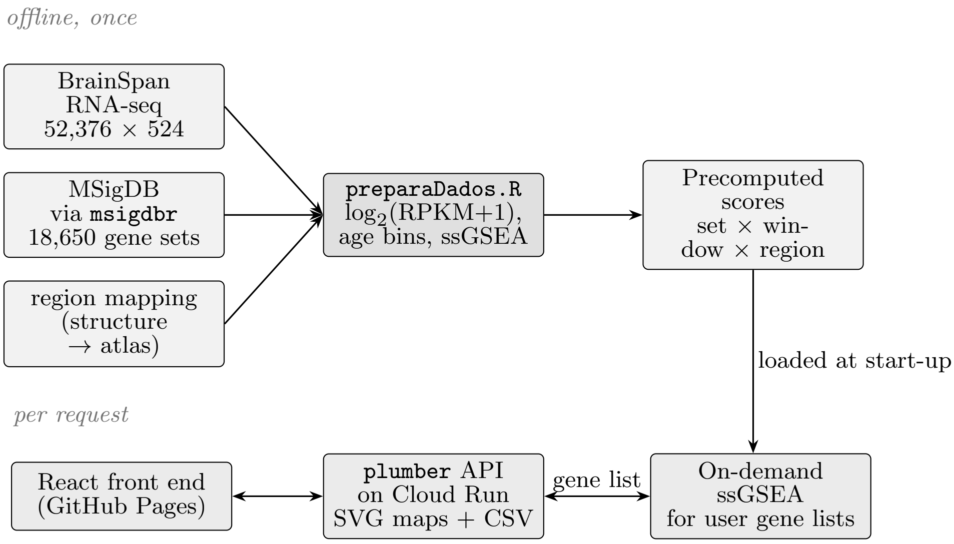

Fig. 1 VEGABrain architecture. Everything to the left of the precomputed objects runs once, offline; everything to the right runs per request. Ontology queries are served by filtering the precomputed score table, whereas a user-submitted gene list triggers a fresh ssGSEA run over all 524 samples.

3 Results: the VEGABrain tool

VEGABrain opens on a single working page (Figure 2) with a control column on the left and the rendered map on the right. Two switches apply to every mode: the region granularity toggle (micro/macro, Section 2.3) and the query mode. All three modes produce the same four-panel, five-window figure, so a user can move between them without relearning the display.

3.1 Mode 1: single gene

Selecting *gene* mode presents a searchable list of every HGNC symbol present in the matrix. Choosing one renders its mean $\log_2(\mathrm{RPKM}+1)$ per region and developmental window on a fixed colour scale bounded at 6, so that maps for different genes are directly comparable rather than each being auto-scaled to its own range. Figure 3 shows *SOX10*, an oligodendrocyte lineage marker, whose expression rises through the perinatal and postnatal windows as myelination proceeds.

3.2 Mode 2: ontology term

Selecting *ontology* mode presents the 18,650 precomputed gene sets. Choosing one returns its mean ssGSEA score per region and window (Figure 4). This is the composite layer described in Section 2.4, and it is what distinguishes VEGABrain from a gene-at-a-time browser: a single map summarises the coordinated behaviour of tens to hundreds of genes, and terms such as `GOBP_FOREBRAIN_GENERATION_OF_NEURONS` resolve into a spatiotemporal pattern that no individual member gene displays cleanly on its own. A companion control lists the genes of the selected set that are present in the matrix, and copies them to the clipboard, so that the composition of any score is inspectable and can be carried into the third mode.

3.3 Mode 3: user-submitted gene list

Pasting a list of symbols—comma-, space- or newline-separated—triggers validation followed by an on-demand ssGSEA run for that set (Figure 5). Symbols absent from the matrix are listed back to the user explicitly, so a map is never quietly computed on a subset of what was submitted. This is the path from an external result to anatomy: the output of a differential expression analysis, a co-expression module, or a disorder risk panel can be mapped across development directly, without the user preparing data or writing code.

3.4 Reference maps and export

Because atlas parcel names are not universally familiar, a reference panel renders the same four views coloured and labelled by region, at either granularity. Every map can be downloaded as SVG and every underlying table as CSV, at whichever granularity is displayed; the CSV contains exactly the values that were painted, so a figure produced in VEGABrain and a statistic computed from its export cannot disagree.

4 Validation and comparison

4.1 Correctness of the scoring pipeline

The precomputed scores served by VEGABrain and a score computed directly from the public matrix must agree, since the precomputation is intended purely as a caching step. We verify this by re-running `GSVA::gsva` with `ssgseaParam` on the same expression matrix for a random sample of gene sets, aggregating the result by developmental window and region exactly as the offline pipeline does, and comparing against the served values.

Across [**TODO:** n] randomly drawn gene sets the served and recomputed per-region scores agreed to a Pearson correlation of [**TODO:** r] with a maximum absolute difference of [**TODO:** Δ], consistent with floating-point accumulation order alone. The on-demand path is validated the same way: submitting the member genes of a precomputed set through the gene-list endpoint reproduces that set's stored map to [**TODO:** Δ].

Table 1 Data underlying VEGABrain.

Property	Value
Expression source	BrainSpan RNA-Seq, Gencode v10 RPKM, gene-summarised
Matrix dimensions	52,376 genes × 524 samples
Transformation	$\log_2(\mathrm{RPKM} + 1)$; duplicate symbols dropped
Developmental windows	5 (see below)
1st trimester	5 donors
2nd trimester	10 donors
3rd trimester	5 donors
Infant	8 donors
Adult	14 donors
Atlas regions (micro)	42 (Desikan–Killiany cortical + `aseg` subcortical)
Anatomical groups (macro)	6 (frontal, parietal, temporal, occipital, subcortical, cerebellum)
Views per map	4 (DK lateral, DK medial, `aseg` sagittal, `aseg` coronal)
Gene sets scored	18,650 total
GO biological process	7,538
Human Phenotype Ontology	5,793
GO molecular function	1,872
Reactome	1,839
GO cellular component	1,080
BioCarta	292
KEGG (legacy)	186
MSigDB hallmark	50
Aggregated score rows	3,188,466 (gene set × window × region)

Table 2 VEGABrain API endpoints. All are public and unauthenticated at https://brainmap-api-5kofy56taq-rj.a.run.app.

Endpoint	Method	Returns
`/list_genes`	GET	All HGNC symbols present in the matrix
`/list_ontologies`	GET	All 18,650 precomputed gene set names
`/plot_brain`	GET	SVG map of one gene's expression
`/plot_ontology`	GET	SVG map of one gene set's ssGSEA score
`/plot_genelist`	POST	SVG map for a submitted gene list (ssGSEA run on demand)
`/data_brain`	GET	CSV of the values behind `/plot_brain`
`/data_ontology`	GET	CSV of the values behind `/plot_ontology`
`/data_genelist`	POST	CSV of the values behind `/plot_genelist`
`/validate_genelist`	POST	Submitted symbols split into matched and unmatched
`/genes_da_via`	GET	Genes of a given set that are present in the matrix
`/plot_reference`	GET	SVG reference map labelled by region name

Every plotting and data endpoint accepts `escala=micro` or `escala=macro` to select region granularity.

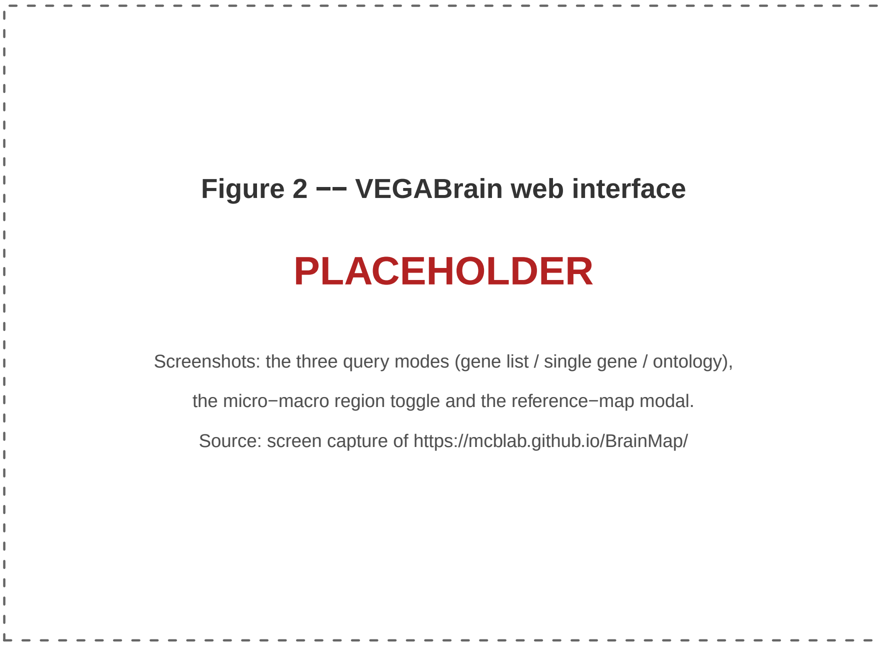

Fig. 2 The VEGABrain interface. Left: query mode selector (gene list, single gene, ontology), the micro/macro granularity toggle, and the search or paste control for the active mode. Right: the rendered four-panel map with its colour legend, the SVG and CSV export buttons, and the reference-map control.

4.2 Biological face validity

Correct arithmetic is not sufficient; the maps must also recover developmental patterns that are independently established. We examined three cases whose expected behaviour is not in dispute.

Oligodendrocyte and myelination markers (*SOX10*, *MBP*, *PLP1*) should be low prenatally and rise sharply in the infant and adult windows; [**TODO: report the observed per-window means**]. GABAergic interneuron markers (*GAD1*, *GAD2*, *SLC32A1*) should be present from the second trimester onward with subcortical and then cortical distribution; [**TODO**]. Neurogenesis-associated ontology terms should peak in the prenatal windows and decline postnatally; [**TODO**]. Where the composite score is informative is precisely here—the individual genes of a neurogenesis term are noisy across 42 sparsely sampled regions, whereas the ssGSEA summary of the term is not.

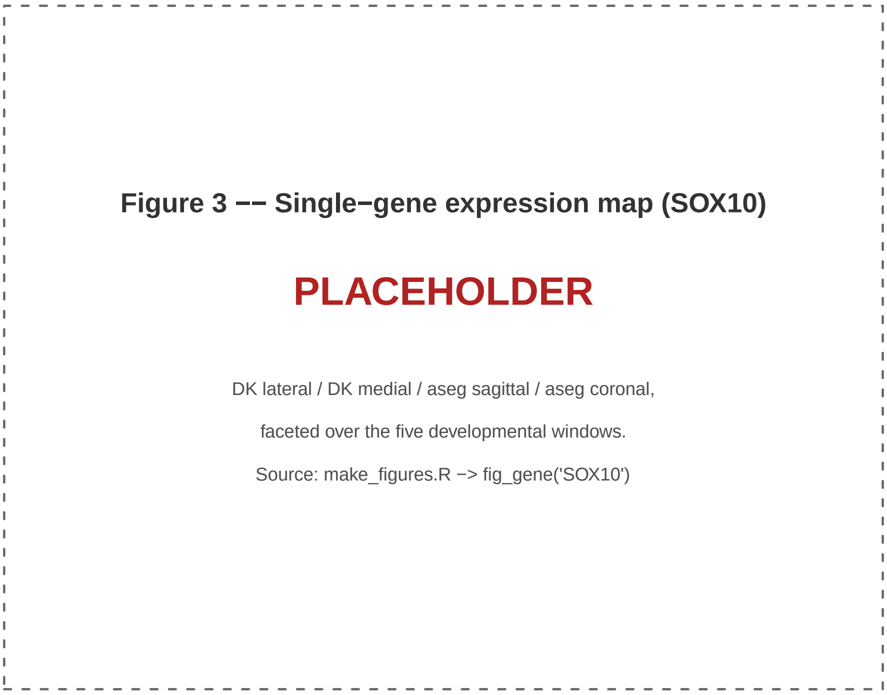

Fig. 3 Single-gene mode. Mean $\log_2(\mathrm{RPKM}+1)$ for *SOX10* across the five developmental windows (columns) in four views (rows): right lateral cortex and right medial cortex on the Desikan–Killiany parcellation, and subcortical structures in sagittal and coronal section on the `aseg` segmentation. Grey indicates regions with no BrainSpan sample in that window.

4.3 Performance

VEGABrain scales to zero when idle, so response time has two regimes. A cold start—container launch plus loading the precomputed objects—was measured at 24.8 s for the first request after an idle period; the front end absorbs this by retrying the initial listing calls while displaying a loading state. Warm requests are dominated by figure rendering for the precomputed modes and by the enrichment run itself for the gene-list mode.

4.4 Comparison with existing tools

Table 4 positions VEGABrain against the tools a researcher would otherwise reach for. The comparison is deliberately along axes of *what the user must supply*, not of raw capability: each of these tools is stronger than VEGABrain in its own domain, and the point is that none of them closes the particular path VEGABrain is built for—from a gene set to a scored, parcellated map of the developing brain, in a browser.

Figure 4 –– Ontology composite ssGSEA map

PLACEHOLDER

GOBP_FOREBRAIN_GENERATION_OF_NEURONS, micro and macro scales.

Headline figure for the per−region enrichment contribution.

Source: make_figures.R −> fig_ontology()

Fig. 4 Ontology mode: composite ssGSEA score for `GOBP_FOREBRAIN_GENERATION_OF_NEURONS` across development. The score summarises every gene in the term into one value per region and window; the same term can be viewed at micro granularity (42 atlas regions) or macro granularity (six anatomical groupings).

Table 3 Response times of the deployed service. Cold start is measured; warm figures are pending a scripted benchmark.

Request	Cold	Warm (median)
`/list_ontologies`	24.8 s	[TODO]
`/plot_brain` (single gene)	—	[TODO]
`/plot_ontology` (precomputed set)	—	[TODO]
`/plot_genelist` (on-demand ssGSEA)	—	[TODO]

The programmatic tools (Mowinckel and Vidal-Piñeiro 2020; Markello et al. 2021, 2022) render brain data beautifully and reproducibly, and VEGABrain is built on one of them, but they take a value per region as *input*: the user must still obtain, preprocess and map the developmental expression data themselves. `cerebroViz` (Bahl et al. 2017) is the closest of them in intent, having been written for spatiotemporal brain data specifically and shipping region vocabularies for BrainSpan, GTEx and

Fig. 5 User-submitted gene-list mode. ssGSEA computed on demand for a GABAergic marker set (*GAD1*, *GAD2*, *SLC32A1*, *GABRB3*) and mapped across development, illustrating the path from an arbitrary gene list to a spatiotemporal map without local installation or coding.

Roadmap Epigenomics; it nonetheless remains a drawing layer over a matrix the user provides, carries no expression data of its own, and computes no enrichment. The same qualification applies to the code-first route as a whole. Chopra et al. (2023) argue for it persuasively and lower its cost with a survey of the available packages and a generator for template plotting code, but a guide to writing the code does not remove the prerequisites the code still has: a working R or Python environment, a local copy of the expression matrix, and a mapping from tissue nomenclature onto a parcellation.

Among hosted resources, BrainScope (Huisman et al. 2017) is the nearest neighbour to VEGABrain, and the only other entry in Table 4 that is at once web-based, developmental and willing to take a user's own gene set. The abstraction differs, and with it the question answered. BrainScope arranges the Allen adult and developing atlases as linked t-SNE maps of genes and samples, localising samples anatomically through choropleths over selected slices, and summarises a submitted set by its mean expression while delegating enrichment to Enrichr and ToppGene. VEGABrain asks the narrower question of where a set sits in a named parcellation, window by window, and answers it with a rank-based composite score rather than a mean, drawn on the

Table 4 VEGABrain relative to comparable tools. ✓ = supported, — = not supported or out of scope.

	Developmental axis	Anatomical rendering	Gene-set scoring	Custom list input	No install required
VEGABrain (this work)	✓	✓	✓	✓	✓
BrainSpan portal (BrainSpan 2011)	✓	—	—	—	✓
Allen browsers (Lau et al. 2008; Sunkin et al. 2012)	—	✓	—	—	✓
BrainScope (Huisman et al. 2017)	✓	✓	—	✓	✓
SMDB (Cao et al. 2023)	—	✓	—	—	✓
`ggseg` (Mowinckel and Vidal-Piñeiro 2020)	—	✓	—	✓	—
`neuromaps` (Markello et al. 2022)	—	✓	—	✓	—
`abagen` (Markello et al. 2021)	—	✓	—	✓	—
`cerebroViz` (Bahl et al. 2017)	✓	✓	—	✓	—
Programmatic plotting guides (Chopra et al. 2023)	—	✓	—	✓	—
Enrichr (Chen et al. 2013; Kuleshov et al. 2016)	—	—	✓	✓	✓
`VoxHunt` (Fleck et al. 2021)	✓	✓	—	✓	—

"Developmental axis" means the tool natively presents variation across prenatal-to-postnatal time. "Anatomical rendering" means values are drawn onto a picture of the brain, in whatever form the tool adopts—a parcellated surface, a slice choropleth or a three-dimensional point cloud. "Gene-set scoring" means it produces a composite per-sample score for a set of genes rather than plotting genes individually; a tool that displays the mean expression of a set, or that hands the set to an external enrichment server, does not qualify. "Custom list input" means the user may supply their own genes or their own per-region values.

Desikan–Killiany and `aseg` atlases that the imaging literature already uses. SMDB (Cao et al. 2023) is likewise hosted and likewise anatomical, but for a different data type and species: it browses spatial transcriptomics sections against Allen's mouse brain coordinate framework and reconstructs morphology in three dimensions, with no human developmental axis and no gene-set layer.

Enrichr (Chen et al. 2013; Kuleshov et al. 2016) performs the gene-set reasoning but returns ranked term tables, with no anatomical or developmental dimension. `VoxHunt` (Fleck et al. 2021) maps expression signatures onto developmental reference atlases, primarily to assign identity to *in vitro* organoid samples, and is an R package rather than a hosted service. VEGABrain occupies the intersection the others leave open: hosted, developmental, parcellated, and gene-set *scored*.

5 Discussion

VEGABrain lowers the cost of asking a spatial question of the developing human transcriptome from a small project to a few seconds in a browser. That change is logistical rather than methodological, but it is the change that determines whether a clinician or an experimental biologist ever looks at these data anatomically. The composite enrichment layer extends the same reasoning to the unit that modern analyses actually produce—a gene set—and the on-demand mode means a user's own list is a first-class input rather than something the tool's authors had to anticipate.

Several limitations bound the interpretation of any map produced here, and the interface is designed to keep them visible rather than to smooth them over.

Donor support is small and uneven.

Each developmental window rests on 5 to 14 donors, and not every structure is sampled in every window. Donor counts are printed in every facet label and unsampled regions are drawn in grey rather than interpolated, but a difference between two regions in a single window should be read as descriptive, not as a test.

The data are bulk tissue.

BrainSpan samples are dissected tissue, so a regional value reflects both cell-intrinsic expression and cell-type composition. A rise in an oligodendrocyte programme across development is as consistent with more oligodendrocytes as with more transcription per cell. Single-cell atlases resolve this and are the natural next data source.

The atlas mapping is an approximation.

BrainSpan dissections and Desikan–Killiany parcels do not correspond one-to-one; several parcels share a BrainSpan structure, and the macro scale pools further. The mapping is published as a single CSV precisely so that it can be inspected, criticised and revised rather than trusted implicitly.

ssGSEA scores are within-sample ranks.

Equation 1 yields a value that is comparable across samples for a fixed gene set, but not across gene sets of different sizes and not on an absolute scale. Colour scales for ontology maps are therefore free rather than fixed, and a score should be read as a spatiotemporal pattern, not as a magnitude.

Scaling to zero trades latency for cost.

The first request after an idle period pays a container cold start. This is a deliberate choice that keeps a public, unfunded service sustainable; the front end masks it, but a user issuing a single query after a quiet period will wait.

Future work follows directly from these bounds: incorporating single-cell and single-nucleus developmental atlases, adding statistical contrast between windows or regions rather than descriptive display alone, allowing user-supplied parcellations, and archiving each release with a citable DOI so that a figure can be tied to an exact version of the data and code.

6 Conclusion

VEGABrain makes the developmental human brain transcriptome visible without installation, coding or data preparation, and adds a composite per-region enrichment layer over 18,650 ontology gene sets together with on-demand scoring of arbitrary user gene lists. Maps and their underlying tables are exportable, the structure-to-atlas mapping is published rather than implied, and the whole stack is open source and containerised. The tool is available at https://mcblab.github.io/BrainMap/.

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Declarations

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Data availability. The expression data analysed here are the BrainSpan Developmental Transcriptome RNA-Seq Gencode v10 summarised-to-genes release, publicly available at https://www.brainspan.org. Gene sets were obtained from MSigDB through the `msigdbr` R package. The derived per-region, per-window score tables served by VEGABrain are downloadable as CSV directly from the application.

Code availability. VEGABrain is open source at https://github.com/MCBLab/BrainMap. The application is deployed at https://mcblab.github.io/BrainMap/ with the API at https://brainmap-api-5kofy56taq-rj.a.run.app.

Author contributions. [TODO]

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